Part 2. Word problems

Example 3. Air enters a spherical balloon at 12π cm^3/s. How fast is its radius increasing when the radius is 2 cm?

Practice 2. A cube's volume decreases at 12 m^3/s. Find the rate of change of its side length when the side is 2 m.

Example 4. A rectangle's length increases at 3 cm/s and its width decreases at 1 cm/s. Find the rate of change of its area when its length is 8 cm and its width is 5 cm.

Practice 3. A triangle's base decreases at 2 cm/min and its height increases at 3 cm/min. Find the rate of change of its area when the base is 8 cm and the height is 6 cm.

Part 3. More challenging models

Example 5. A 10-ft ladder rests against a vertical wall. Its bottom moves away from the wall at 2 ft/s. (a) How fast and in what direction does the top move when the bottom is 6 ft from the wall? (b) At this position, if the top instead moves downward at 3 ft/s, how fast does the bottom move?

Practice 4. A rope connects a boat's bow to a dock pulley 3 m above it. The bow stays at a fixed height, and the rope is pulled in at 1 m/s. How fast does the boat approach the dock when the horizontal distance from bow to pulley is 4 m?

Example 6. An inverted conical tank is 6 m tall with top radius 2 m. Water enters at π m^3/min. (a) Express the water volume in terms of its depth h. (b) How fast does the water level rise when $h = 3$ m? (c) At this depth, how would the rate change if water instead drained at π m^3/min?

Practice 5. A 2-m-tall person walks away from a 6-m-tall streetlight at 2 m/s on level ground. How fast are (a) the length of the person's shadow and (b) the distance from the streetlight to the tip of the shadow increasing?